Average Velocity and Speed

1. A particle moves according to the equation $x = t^2 - 4$, where x is the displacement in metres from the origin O at time t seconds after time zero.
   a. Copy and complete the table below of values of the displacement at certain times.

t	0	1	2	3
x				

   b. Hence find the average velocity:
      i. During the first second,
      ii. During the first two seconds,
      iii. During the first three seconds,
      iv. During the third second.
   c. Sketch the displacement-time graph, and add the chords corresponding to the average velocities calculated in part b.

2. A particle moves according to the equation $x = 2\sqrt{t}$, for $t \geq 0$, where distance is in centimetres and time is in seconds.
   a. Copy and complete the table below.

t					
x	0	2	4	6	8

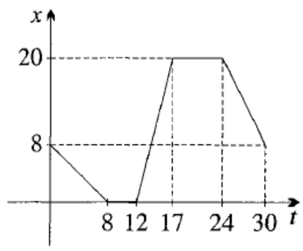
   b. Hence find the average velocity as the particle moves:
      i. From $x = 0$ to $x = 2$
      ii. From $x = 2$ to $x = 4$
      iii. From $x = 4$ to $x = 6$
      iv. From $x = 0$ to $x = 6$.
   c. Sketch the displacement-time graph, and add the chords corresponding to the average velocities calculated in part b. What does the equality of the answer to parts ii. and iv. of part b. tell you about the corresponding chords?

3. A particle moves according to the equation $x = 4t - t^2$, where distance is in metres and time is in seconds.
   a. Copy and complete the table below.

t	0	1	2	3	4
x					

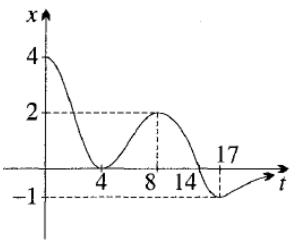
   b. Hence find the average velocity as the particle moves:
      i. From $t = 0$ to $t = 2$,
      ii. From $t = 2$ to $t = 4$,
      iii. From $t = 0$ to $t = 4$.
   c. Sketch the displacement-time graph, and add the chords corresponding to the average velocities calculated in part b.
   d. Find the total distance travelled during the first 4 seconds, and the average speeds over the time intervals specified in part b.

4. Eleni is practicing reversing in her driveway. Starting 8 metres from the gate, she reverses to the gate, and pauses. Then she drives forward 20 metres, and pauses. Then she reverses to her starting point. The graph to the right shows her distance x metres from the front gate after t seconds.

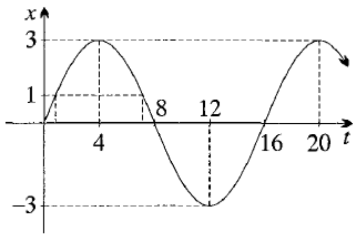


   a. What is her velocity:
      i. During the first 8 seconds,
      ii. While she is driving forwards,
      iii. While she is reversing the second time?
   b. Find the total distance travelled, and the average speed, over the 30 seconds.
   c. Find the change in displacement, and the average velocity, over the 30 seconds.
   d. Find her average speed if she had not paused at the gate and at the garage.

5. Michael the mailman rides $1\ km$ up a hill at a constant speed of $10\ km/hr$, and then rides $1\ km$ down the other side of the hill at a constant speed of $30\ km/hr$.
    a. How many minutes does he take to ride:
        i. Up the hill,
        ii. Down the hill?
    b. Draw a displacement-time graph, with the time axis in minutes.
    c. What is his average speed over the total $2\ km$ journey?
    d. What is the average of the speeds up and down the hill?
6. Sadie the snail is crawling up a 6 metre high wall. She takes an hour to crawl up 3 metres, then falls asleep for an hour and slids down 2 metres, repeating the cycle until she reaches the top of the wall.
    a. Sketch the displacement-time graph.
    b. How long does Sadie take to reach the top?
    c. What is her average speed?
    d. Which places does she visit exactly three times?
7. A girl is leaning over a bridge 4 metres above the water, playing with a weight on the end of a spring. The diagram graphs the height x in metres of the weight above the water as a function of time t after she first drops it.



    a. How many times is the weight:
        i. At $x = 3$,
        ii. At $x = 1$,
        iii. At $x = -\frac{1}{2}$?
    b. At what times is the weight:
        i. At the water surface,
        ii. Above the water surface?
    c. How far above the water does it rise again after it first touches the water, and when does it reach this greatest height?
    d. What is the greatest depth under the water, and wen does it occur?
    e. What happens to the weight eventually?
    f. What is its average velocity:
    g. What distance does it travel:
        i. Over the first 4 seconds,
        ii. Over the first 8 seconds.
        iii. Over the first 17 seconds
        iv. Eventually?
    h. What is its average speed over the first
        i. 4,
        ii. 8,
        iii. 17 seconds?
8. A particle is moving according to $x = 3\sin\frac{\pi}{8}t$, in units of centimetres and seconds. Its displacement-time graph is sketched opposite.



    a. Use $T = \frac{2\pi}{n}$ to confirm that the period is 16 seconds.
    b. Find the first two times when the displacement is maximum.
    c. When, during the first 20 seconds, is the particle on the negative side of the origin?
    d. Find the total distance travelled during the first 16 seconds, and the average speed.
    e. i. Find, correct to three significant figures, the first two positive solutions of the trigonometric equation $\sin\frac{\pi}{8}t = \frac{1}{3}$. [Hint: Use radian mode on the calculator.]
        ii. Hence find, correct to three significant figures, the first two times when $x = 1$. Then find the total distance travelled between these two times and the average speed during this time.

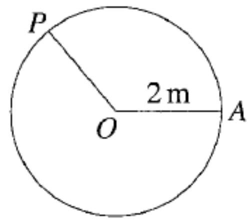
9. A particle moves according to $x = 10\cos\frac{\pi}{12}t$, in units of metres and seconds.
    a. Find the amplitude and period of the motion.
    b. Sketch the displacement-time graph over the first 60 seconds.
    c. What is maximum distance the particle reaches from its initial position, and when, during the first minute, is it there?
    d. How far does the particle move during the first minute, and what is its average speed?
    e. When, during the first minute, is the particle 10 metres from its initial position?
    f. Use the fact that $\cos\frac{\pi}{3} = \frac{1}{2}$ to copy and complete this table of values:

t	4	8	12	16	20	24
x						

    g. From the table, find the average velocity during the first 4 seconds, the second 4 seconds, and the third 4 seconds.
    h. Use the graph and the table of values to find when the particle is more than 15 metres from its initial position.
10. A particle is moving on a horizontal number line according to the equation $x = 4\sin\frac{\pi}{6}t$, in units of metres and seconds.
    a. Sketch the displacement-time graph.
    b. How many times does the particle return to the origin by the end of the first minute?
    c. Find at what times it visits $x = 4$ during the first minute.
    d. Find how far it travels during the first 12 seconds, and its average speed in that time.
    e. Find the values of x when $t = 0$, $t = 1$ and $t = 3$. Hence show that its average speed during the first second is twice its average speed during the next 2 seconds.
11. A balloon rises so that its height h in metres after t minutes is $h = 8000(1 - e^{-0.06t})$.
    a. What height does it start from, and what happens to the height as $t \to \infty$?
    b. Copy and complete the table to the right, correct to the nearest metre.

t	0	10	20	30
h				

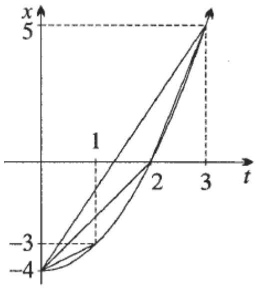
    c. Sketch the displacement-time graph of the motion.
    d. Find the balloon's average velocity during the first 10 minutes, the second 10 minutes and the third 10 minutes, correct to the nearest metre per minute.
    e. Show that the solution of $1 - e^{-0.06t} = 0.99$ is $t = \frac{\log 100}{0.06}$.
    f. Hence find how long (correct to the nearest minute) the balloon takes to reach 99% of its final height.
12. A toy train is travelling anticlockwise on a circular track of radius 2 metres and centre O. At time zero the train is at a point A, and t seconds later it is at the point P distant $x = 4\log(t + 1)$ metres around the track.



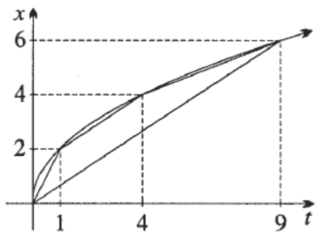
    a. Sketch the graph of x as a function of t.
    b. Find, when $t = 2$, the position of the point P, the average speed from A to P, the size of $\angle AOP$ and the length of the chord AP (in exact form, then correct to four significant figures).
    c. More general, find $\angle AOP$ as a function of t. Hence find, in exact form, and then correct to the nearest second, the first three times when the train returns to A.
    d. Explain whether the train will return to A finitely or infinitely many times.

Answer:

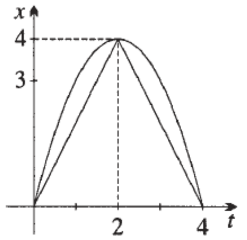
1. a. $x = -4, -3, 0, 5$; b. i. $1\ m/s$, ii. $2\ m/s$, iii. $3\ m/s$, iv. $5\ m/s$



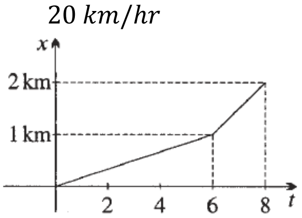
2. a. $t = 0, 1, 4, 9, 16$; b. i. $2\ cm/s$, ii. $\frac{2}{3}\ cm/s$, iii. $\frac{2}{5}\ cm/s$, iv. $\frac{2}{3}\ cm/s$; c. They are parallel.



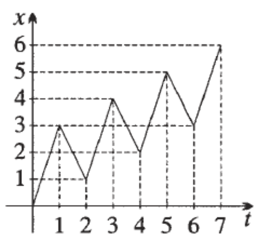
3. a. $x = 0, 3, 4, 3, 0$; b. i. $2\ m/s$, ii. $-2\ m/s$, iii. $0\ m/s$; d. The total distance travelled is 8 metres. All three average speeds are $2\ m/s$.



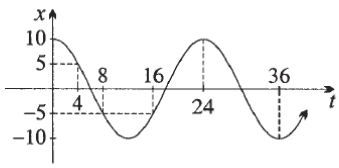
4. a. i. $-1\ m/s$, ii. $4\ m/s$, iii. $-2\ m/s$; b. 40 metres, $1\frac{1}{3}\ m/s$; c. 0 metres, $0\ m/s$; d. $2\frac{2}{19}\ m/s$
5. a. i. 6 minutes, ii. 2 minutes; c. $15\ km/hr$; d. $20\ km/hr$



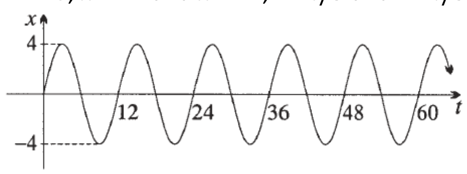
6. b. 7 hours; c. $2\frac{4}{7} m/hr$; d. those between 1 and 2 metres high or between 4 and 5 metres high



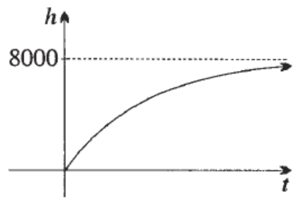
7. a. i. once, ii. three times, iii. Twice; b. i. when $t = 4$ and when $t = 14$, ii. when $0 \leq t < 4$ and when $4 < t < 14$; c. It rises 2 metres, at $t = 8$; d. It sinks 1 metre, at $t = 17$; e. As $t \to \infty$, $x \to 0$, meaning that eventually it ends up at the surface; f. i. $-1\ m/s$, ii. $\frac{1}{2}\ m/s$, iii. $-\frac{1}{3}\ m/s$; g. i. 4 metres, ii. 6 metres, iii. 9 metres, iv. 10 metres; h. i. $1\ m/s$, ii. $\frac{3}{4}\ m/s$, iii. $\frac{9}{17}\ m/s$
8. b. $t = 4$, $t = 20$; c. $8 < t < 16$; d. $12\ cm$, $\frac{3}{4}\ cm/s$; e. i. $t = \frac{8}{\pi}\sin^{-1}\frac{1}{3} \doteqdot 0.865$, and $t = 8 - \frac{8}{\pi}\sin^{-1}\frac{1}{3} \doteqdot 7.13$, ii. 0.865 seconds and 7.13 seconds, 4 metres, $0.638\ m/s$.
9. Amplitude: 10 metres, period: 24 seconds (graph see below). c. the maximum is 20 metres, when $t = 12$, 36 and 60; d. 100 metres, $1\frac{2}{3}\ m/s$; e. It is at $x = 0$ when $t = 6$, 18, 30, 42 and 54; f. 10, 5, −5, −10, −5, 5, 10; g. $-1\frac{1}{4}\ m/s$, $-2\frac{1}{2}\ m/s$, $-1\frac{1}{4}\ m/s$; h. $x = -5$ when $t = 8$ or $t = 16$, $x < -5$ when $8 < t < 16$.



10. a. amplitude: 4 metres, period: 12 seconds (graph see below); b. 10 times; c. $t = 3, 15, 27, 39, 51$; d. It travels 16 metres with average speed $1\frac{1}{3}\ m/s$; e. $x = 0$, $x = 2$ and $x = 4$, $2\ m/s$ and $1\ m/s$



11. a. When $t = 0$, $h = 0$. As $t \to \infty$, $h \to 8000$; b. 0, 3610, 5590, 6678; d. $361\ m/min$, $198\ m/min$, $109\ m/min$; f. 77 minutes.



12. a. see below; b. When $t = 2$, $x = 4\log 3 \doteqdot 4.394$, the average speed is $2\log 3 \doteqdot 2.197\ m/s$, $\angle AOP = 2\log 3 \doteqdot 2.197\ radians$, $AP^2 = 8(1 - \cos(2\log 3))$, $AP \doteqdot 3.562$ metres; c. $\angle AOB = 2\log(t+1)$. The train is at A when $t = e^{\pi} - 1 \doteqdot 22$, when $t = e^{2\pi} - 1 \doteqdot 534$, and when $t = e^{3\pi} - 1 \doteqdot 12391$; d. Since $2\log(t+1) \to \infty$ as $t \to \infty$, the train will return to A infinitely many times.